Cohen Bryant

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EDUCATION

University of Wisconsin–Madison | Madison, WI

Bachelor of Science in Mechanical Engineering | Expected Graduation: May 2030

- **Relevant Coursework:** Calculus II/III, Statics, Introduction to Engineering Design (Completed), Computer-Aided Engineering (SolidWorks), Python/MatLab Programming.

TECHNICAL SKILLS

- **CAD & Simulation:** SolidWorks (3D Assemblies, GD&T, Stress, Vibration & Aerodynamic FEA), FEA/CFD Principles
- **Programming & Tools:** C++, JavaScript, Python, HTML/CSS, Git, Vercel, Arduino IDE, MATLAB
- **Hardware & Electronics:** Arduino Nano ESP32, Motor Controllers, Sensors (Thermocouples, Load Cells), Microcontrollers
- **Fabrication & Machining:** 3D Printing (FDM/SLA), Lathe, Mill, Soldering, Hand Tools, Prototype Assembly

TECHNICAL & COMPUTATIONAL PROJECTS

Transient Nuclear Reactor Xenon-135 Simulator | *Creator & Developer* |

- **Built a web-based computational simulator to model transient Xenon-135 reactivity poisoning and reactor dead-out periods following thermal neutron flux step changes.**
- **Programmed differential decay kinetics for Iodine-135 and Xenon-135 in JavaScript to plot dynamic power shifts, peak poison concentrations, and override windows.**
- **Designed an interactive UI to evaluate reactor control constraints and steady-state equilibrium power transitions.**

Arduino ESP32 Tracked Vehicle & Hardware Controls | *Hardware & Control Lead*

- **Programmed an Arduino Nano ESP32 in C++ to manage motor drive logic, speed control, and hardware signal processing for an autonomous tracked vehicle.**
- **Modeled custom chassis and drivetrain components in SolidWorks, applying tight tolerances for 3D-printed drive sprockets and motor brackets.**

- **Resolved runtime C++ compilation errors, structured code directories, and documented wiring schematics to ensure hardware reliability.**

Custom Launcher & Trajectory Calculator | *Mechanical Design & Analysis*

- **Applied calculus and physics equations to model energy transfer, apex height, and trajectory distance for a mechanical launcher build.**
- **Machined structural support components and adjusted spring tension parameters to align physical test launch data within 5% of theoretical predictions.**

Nuclear Systems Analytics Dashboard (NucInfo) | *Developer & Researcher* |

- **Developed an interactive web analytics platform visualizing nuclear fission/fusion thermodynamics, 93% capacity factor metrics, and lifecycle carbon intensity (12% g CO_2/kWh).**

ME Connect Student Engineering Platform | *Creator & Web Architect* |

- **Architected and deployed a centralized networking platform to connect UW-Madison engineering undergraduates with faculty, mentorship portals, and course resources.**

LEADERSHIP & CAMPUS INVOLVEMENT

AIAA / Liquid Rocketry / Formula SAE Student Team | *Active Member* | Aug 2026 – Present

- Collaborate with upperclassmen engineers on CAD modeling, subsystem fabrication, and test rig setup for upcoming competition builds.

ADDITIONAL EXPERIENCE

Certified Nursing Assistant | *North Central Health Care* | 8/26/2024 - 7/30/2026

- Working in collaborative efforts with team members in a high-stress environment
- Effective communication with peers and coworkers